Generalized Linear Mixed Models

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Part 1: Introduction of Generalized Linear Mixed Models

Motivation I

Recall the basic assumptions of the linear models where $y_1, \ldots, y_m$ are the observed response variable and $\mathbf{x}_1, \ldots, \mathbf{x}_m \in \mathbb{R}^{p\times 1}$ are the individual covariates (here $Z \in \mathbb{R}^{a\times b}$ means that Z is a $(a \times b)$ real-valued matrix):

1. $E[y_i] = \mathbf{x}_i'\boldsymbol{\beta}$ for $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p) \in \mathbb{R}^{p\times 1}$,
2. $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\}$ are non-random,
3. $Var(y_i) = \sigma^2$,
4. $\{y_1, y_2, \ldots, y_m\}$ are independent,
5. $\{y_1, y_2, \ldots, y_m\}$ are normally distributed.

Such situations can be written in short as follows:

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2). \tag{1}$$

Note that the first components of $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_m\}$ are all 1 so that β_1 corresponds to the *intercept* term in the linear models.

Motivation II

Now, let us assume an example of data with hierarchy so that we observe data points for each of the subjects $i \in \{1, \ldots, m\}$ for time $t = 1, \ldots, n_i$. If we adopt the assumptions from the linear models, then we can write them as follows with the modified subscripts:

1. $E[y_{i,t}] = \mathbf{x}'_{i,t}\boldsymbol{\beta}$ for $\beta \in \mathbb{R}^{p \times 1}$,
2. $\{\mathbf{x}_{1,1}, \mathbf{x}_{1,2}, \ldots, \mathbf{x}_{m,n_m}\}$ are non-random,
3. $Var(y_{i,t}) = \sigma^2$,
4. $\{y_{1,1}, y_{1,2}, \ldots, y_{m,n_m}\}$ are independent,
5. $\{y_{1,1}, y_{1,2}, \ldots, y_{m,n_m}\}$ are normally distributed.

What if both of Assumptions 4 and 5 are not valid...? (Note that $\{y_{i,1}, y_{i,2}, \ldots, y_{i,n_i}\}$ are from the same subject i over time)

Motivation III

In this regard, one can consider the following representation:

$$\mathbb{E}[y_{i,t}|u_i] = \mu_{i,t}, \quad \eta(\mu_{i,t}) = \mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i, \tag{2}$$

where u_i is a random variable for $i = 1, \ldots, m$. Note that it is usual to impose either $\mathbb{E}[u_i] = 0$ or $\mathbb{E}[e^{u_i}] = 1$.

Although we may incorporate random effects in various ways (not only as the random intercept), for this section we stick to the form in (2). Again, $y_{i,t}$ and $y_{i,t'}$ are NOT independent!

Examples I

Example (Normal GLMM)

Consider the case where $y_{i,t}|u_i$ follows a normal distribution (which belongs to the exponential families) and $\eta(z) = z$ (identity link). In this case, we have

$$y_{i,t}|u_i \overset{indep}{\sim} \mathcal{N}(\mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i, \sigma^2), \quad \mathbb{E}[u_i] = 0,$$

which is indeed equivalent to $y_{i,t} = \mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i + \epsilon_{i,t}$, $\epsilon_{i,t} \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$. Therefore, one can confirm that the GLMMs are natural extensions of the LMMs.

Examples II

Example (Bernoulli GLMM - logistic regression)

Consider the case where $y_{i,t}|u_i$ follows a Bernoulli distribution (a special case of the binomial distribution with size 1) and $\eta(z) = \log\left(\frac{z}{1-z}\right)$ (logit link). In this case, we have

$$y_{i,t}|u_i \overset{indep}{\sim} \mathcal{B}er\left(\frac{e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}}{1+e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}}\right), \quad \mathbb{E}[u_i] = 0,$$

so that $\log\left(\frac{E[y_{i,t}|u_i]}{1-E[y_{i,t}|u_i]}\right) = \mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i$, and

$\mathbb{P}(y_{i,t} = 1|u_i) = \mathbb{E}[y_{i,t}|u_i] = \frac{e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}}{1+e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}}$.

Examples III

Example (Poisson GLMM)

Consider the case where $y_{i,t}|u_i$ follows a Poisson distribution (which belongs to the exponential families) and $\eta(z) = \log z$ (log link). In this case, we have

$$y_{i,t}|u_i \overset{indep}{\sim} \mathcal{P}(e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}), \quad \mathbb{E}[e^{u_i}] = 1,$$

so that $\log \mathbb{E}[y_{i,t}|u_i] = \mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i$ and $\mathbb{E}[y_{i,t}|u_i] = e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}$.

While it is not strictly required, we often assume that $\Theta_i := e^{u_i} \sim \mathcal{G}(r, 1/r)$ so that $\mathbb{E}[\Theta_i] = 1$ and $\mathbb{V}ar(\Theta_i) = 1/r$.

Examples IV

Example (Gamma GLMM)

Consider the case where $y_{i,t}|u_i$ follows a gamma distribution and $\eta(z) = \log z$ (log link). In this case, we have

$$y_{i,t}|u_i \overset{indep}{\sim} \mathcal{G}(1/\psi, \psi \cdot e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}), \quad \mathbb{E}[e^{u_i}] = 1,$$

so that $\log \mathbb{E}[y_{i,t}|u_i] = \mathbf{x}'_{i,t}\boldsymbol{\beta} + u_i$ and $\mathbb{E}[y_{i,t}|u_i] = e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}+u_i}$.

While it is not strictly required, we often assume that $\Theta_i := e^{u_i} \sim \mathcal{IG}(r+1, r)$ so that $\mathbb{E}[\Theta_i] = 1$ and $\mathbb{V}ar(\Theta_i) = \frac{r}{r-1}$.

Estimation I

- Due to the common random effect u_i that affects all observations $y_{i,1}, \ldots, y_{i,n_i}$ simultaneously, $y_{i,1}, \ldots, y_{i,n_i}$ are not marginally independent.
- So in the maximum likelihood estimation of $\boldsymbol{\beta}$, we need to consider the following joint densities $h(y_{i,1}, \ldots, y_{i,n_i})$, which is given as follows:

$$h(y_{i,1}, \ldots, y_{i,n_i}) = \int \left[\prod_{t=1}^{n_i} f(y_{i,t}|u_i) \right] \pi(u_i) du_i,$$

where $\pi(u_i)$ is the PDF of u_i.

Estimation II

- Therefore, the loglikelihood is given as follows:

$$\begin{aligned}\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta}) \\ &= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | u_i) \right] \pi(u_i) du_i,\end{aligned}$$

and the following score equation

$$S(\boldsymbol{\beta}) = \frac{\partial}{\partial \boldsymbol{\beta}} \ell(\boldsymbol{\beta}) = 0$$

is not easy to be explicitly obtained due to the integrals inside the summation.

Estimation III

In general, $\ell(\boldsymbol{\beta})$ is not explicitly expressed without the integrals so one can consider the following ways for finding the MLE of $\boldsymbol{\beta}$:

- EM algorithms: 'pretend' that you have complete observations of u_i, and iterate to obtain the $\hat{\boldsymbol{\beta}}$,
- The adaptive Gauss-Hermite approximation: approximate the integral as a finite sum and then evalute the likelihood.

If we assume u_i follows a normal distribution, then we can use the adaptive Gauss-Hermite approximation to find $\hat{\boldsymbol{\beta}}$ in `R` using a package (e.g., `glmer`).

Estimation IV

For the remainder of this chapter, we focus on some special cases where we can obtain an explicit form of $\ell(\boldsymbol{\beta})$ with special combinations of $y_{i,t}|u_i$ and u_i (or $y_{i,t}|\Theta_i$ and Θ_i).

- Normal-Normal GLMM:

$$
\begin{aligned}
y_{i,t}|u_i &\overset{indep}{\sim} \mathcal{N}(\mathbf{x}_{i,t}'\boldsymbol{\beta} + u_i, \sigma^2) \text{ for } t = 1, \ldots, n_i, \\
u_i &\overset{indep}{\sim} \mathcal{N}(0, \phi_u) \text{ for } i = 1, \ldots, m. \\
\Longrightarrow \boldsymbol{y}_i &\sim \mathcal{N}\left(\mathbf{x}_i\boldsymbol{\beta},\, \phi_u \cdot J_{n_i} + \sigma^2 \cdot I_{n_i}\right),
\end{aligned}
$$

where $\boldsymbol{y}_i = (y_{i,1}, \ldots, y_{i,n_i})' \in \mathbb{R}^{n_i \times 1}$.

Estimation V

- Poisson-gamma GLMM:

$$y_{i,t}|\Theta_i \overset{indep}{\sim} \mathcal{P}(\Theta_i \cdot e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}}) \text{ for } t = 1, \ldots, n_i,$$

$$\Theta_i \overset{indep}{\sim} \mathcal{G}(r, 1/r) \text{ for } i = 1, \ldots, m.$$

$$\Longrightarrow \boldsymbol{y}_i \text{ follows a } \textit{multivariate negative binomial} \text{ distribution,}$$

where $\boldsymbol{y}_i = (y_{i,1}, \ldots, y_{i,n_i})' \in \mathbb{R}^{n_i \times 1}$.

Estimation VI

- Gamma-inverse gamma GLMM:

$$y_{i,t}|\Theta_i \overset{indep}{\sim} \mathcal{G}(1/\psi, \psi \cdot \Theta_i \cdot e^{\mathbf{x}_{i,t}'\boldsymbol{\beta}}) \text{ for } t = 1, \ldots, n_i,$$

$$\Theta_i \overset{indep}{\sim} \mathcal{IG}(r+1, r) \text{ for } i = 1, \ldots, m.$$

$$\Longrightarrow \boldsymbol{y}_i \text{ follows a } \textit{multivariate generalized Pareto} \text{ distribution,}$$

where $\boldsymbol{y}_i = (y_{i,1}, \ldots, y_{i,n_i})' \in \mathbb{R}^{n_i \times 1}$.

Poisson-gamma GLMM I

Theorem (Fundamental Theorem of Calculus)

Suppose $\frac{d}{dt}F(t) = f(t)$, *then*

$$\int_a^b f(t)dt = F(b) - F(a).$$

Poisson-gamma GLMM II

Theorem (Fundamental Theorem of the Loss Models(?))

If $\Gamma(r)$ *is defined in the following way,*

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x} dx,$$

then $\Gamma(r) = (r-1)\Gamma(r-1)$.
Especially, if r *is a natural number, then* $\Gamma(r) = (r-1)!$.

Poisson-gamma GLMM III

Now, we are ready to evaluate the joint PMF of $(y_{i,1}, \ldots, y_{i,n_i})$ in the case of Poisson-gamma GLMM, which is given as follows:

$$h(y_{i,1}, \ldots, y_{i,n_i}) = \int \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i,$$

Recall that in the case of Poisson-gamma GLMM, we have

$$f(y_{i,t}|\theta_i) = \frac{(\theta_i \nu_{i,t})^{y_{i,t}} e^{-\theta_i \nu_{i,t}}}{y_{i,t}!} \text{ for } t = 1, \ldots, n_i,$$

$$\pi(\theta_i) = \frac{r^r}{\Gamma(r)} \theta_i^{r-1} e^{-r\theta_i},$$

where $\nu_{i,t} = \exp\left(\mathbf{x}'_{i,t}\boldsymbol{\beta}\right)$.
Can you see that $\mathbb{E}[\Theta_i] = \int_0^\infty \theta_i \pi(\theta_i) = 1$?

Multivariate negative binomial distribution

Let us first define $\nu_{i,\bullet} = \sum_{t=1}^{n_i} \nu_{i,t}$ and $y_{i,\bullet} = \sum_{t=1}^{n_i} y_{i,t}$.

$$h(y_{i,1}, \ldots, y_{i,n_i})$$
$$= \int_0^{\infty} \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i$$

Multivariate negative binomial distribution

Let us first define $\nu_{i,\bullet} = \sum_{t=1}^{n_i} \nu_{i,t}$ and $y_{i,\bullet} = \sum_{t=1}^{n_i} y_{i,t}$.

$$
\begin{aligned}
& h(y_{i,1}, \ldots, y_{i,n_i}) \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} \frac{(\theta_i \nu_{i,t})^{y_{i,t}} e^{-\theta_i \nu_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \theta_i^{r-1} e^{-r\theta_i} d\theta_i,
\end{aligned}
$$

Multivariate negative binomial distribution

Let us first define $\nu_{i,\bullet} = \sum_{t=1}^{n_i} \nu_{i,t}$ and $y_{i,\bullet} = \sum_{t=1}^{n_i} y_{i,t}$.

$$
\begin{aligned}
& h(y_{i,1}, \ldots, y_{i,n_i}) \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} \frac{(\theta_i \nu_{i,t})^{y_{i,t}} e^{-\theta_i \nu_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \theta_i^{r-1} e^{-r\theta_i} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \left[\prod_{t=1}^{n_i} (\theta_i)^{y_{i,t}} e^{-\theta_i \nu_{i,t}} \right] \theta_i^{r-1} e^{-r\theta_i} d\theta_i,
\end{aligned}
$$

Multivariate negative binomial distribution

Let us first define $\nu_{i,\bullet} = \sum_{t=1}^{n_i} \nu_{i,t}$ and $y_{i,\bullet} = \sum_{t=1}^{n_i} y_{i,t}$.

$$
\begin{aligned}
& h(y_{i,1}, \ldots, y_{i,n_i}) \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} \frac{(\theta_i \nu_{i,t})^{y_{i,t}} e^{-\theta_i \nu_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \theta_i^{r-1} e^{-r\theta_i} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \left[\prod_{t=1}^{n_i} (\theta_i)^{y_{i,t}} e^{-\theta_i \nu_{i,t}} \right] \theta_i^{r-1} e^{-r\theta_i} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{\sum_{t=1}^{n_i} y_{i,t} + r - 1} e^{-\theta_i (\sum_{t=1}^{n_i} \nu_{i,t} + r)} d\theta_i,
\end{aligned}
$$

Multivariate negative binomial distribution

Let us first define $\nu_{i,\bullet} = \sum_{t=1}^{n_i} \nu_{i,t}$ and $y_{i,\bullet} = \sum_{t=1}^{n_i} y_{i,t}$.

$$
\begin{aligned}
& h(y_{i,1}, \ldots, y_{i,n_i}) \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} f(y_{i,t}|\theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \int_0^\infty \left[\prod_{t=1}^{n_i} \frac{(\theta_i \nu_{i,t})^{y_{i,t}} e^{-\theta_i \nu_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \theta_i^{r-1} e^{-r\theta_i} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \left[\prod_{t=1}^{n_i} (\theta_i)^{y_{i,t}} e^{-\theta_i \nu_{i,t}} \right] \theta_i^{r-1} e^{-r\theta_i} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{\sum_{t=1}^{n_i} y_{i,t} + r - 1} e^{-\theta_i (\sum_{t=1}^{n_i} \nu_{i,t} + r)} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{y_{i,\bullet} + r - 1} e^{-\theta_i (\nu_{i,\bullet} + r)} d\theta_i,
\end{aligned}
$$

Multivariate negative binomial distribution

By letting $u = \theta_i(\nu_{i,\bullet} + r)$, we have

$$= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{y_{i,\bullet}+r-1} e^{-\theta_i(\nu_{i,\bullet}+r)} d\theta_i,$$

Multivariate negative binomial distribution

By letting $u = \theta_i(\nu_{i,\bullet} + r)$, we have

$$= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{y_{i,\bullet}+r-1} e^{-\theta_i(\nu_{i,\bullet}+r)} d\theta_i,$$

$$= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \frac{1}{(\nu_{i,\bullet}+r)^{y_{i,\bullet}+r}} \int_0^\infty u^{y_{i,\bullet}+r-1} e^{-u} du,$$

Multivariate negative binomial distribution

By letting $u = \theta_i(\nu_{i,\bullet} + r)$, we have

$$
\begin{aligned}
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \int_0^\infty \theta_i^{y_{i,\bullet}+r-1} e^{-\theta_i(\nu_{i,\bullet}+r)} d\theta_i, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \frac{1}{(\nu_{i,\bullet}+r)^{y_{i,\bullet}+r}} \int_0^\infty u^{y_{i,\bullet}+r-1} e^{-u} du, \\
&= \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!}\right] \frac{r^r}{\Gamma(r)} \times \frac{\Gamma(y_{i,\bullet}+r)}{(\nu_{i,\bullet}+r)^{y_{i,\bullet}+r}}!
\end{aligned}
$$

After such a tedious calculation, we finally removed the integral from the joint PMF $h(y_{i,1}, \ldots, y_{i,n_i})$ and now ready to (eventually) write out $\ell(\boldsymbol{\beta})$ explicitly.

Poisson-gamma GLMM full loglikelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta})$$

Poisson-gamma GLMM full loglikelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta})$$
$$= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | \theta_i) \right] \pi(\theta_i) d\theta_i$$

Poisson-gamma GLMM full loglikelihood

$$
\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta}) \\
&= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | \theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t} - \log y_{i,t}!) + r \log r \right.
\end{aligned}
$$

Poisson-gamma GLMM full loglikelihood

$$
\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta}) \\
&= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | \theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t} - \log y_{i,t}!) + r \log r \right. \\
&\quad \left. - \log \Gamma(r) + \log \Gamma(y_{i,\bullet} + r) - (y_{i,\bullet} + r) \log(\nu_{i,\bullet} + r) \right],
\end{aligned}
$$

Poisson-gamma GLMM full loglikelihood

$$
\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta}) \\
&= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | \theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t} - \log y_{i,t}!) + r \log r \right. \\
&\quad \left. - \log \Gamma(r) + \log \Gamma(y_{i,\bullet} + r) - (y_{i,\bullet} + r) \log(\nu_{i,\bullet} + r) \right], \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t})) - (y_{i,\bullet} + r) \log(\nu_{i,\bullet} + r) \right]
\end{aligned}
$$

Poisson-gamma GLMM full loglikelihood

$$
\begin{aligned}
\ell(\boldsymbol{\beta}) &= \sum_{i=1}^{m} \log h(y_{i,1}, \ldots, y_{i,n_i}; \mathbf{x}_{i,1}, \ldots, \mathbf{x}_{i,n_i} | \boldsymbol{\beta}) \\
&= \sum_{i=1}^{m} \log \int \left[\prod_{t=1}^{n_i} f(y_{i,t} | \theta_i) \right] \pi(\theta_i) d\theta_i \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t} - \log y_{i,t}!) + r \log r \right. \\
&\quad \left. - \log \Gamma(r) + \log \Gamma(y_{i,\bullet} + r) - (y_{i,\bullet} + r) \log(\nu_{i,\bullet} + r) \right], \\
&= \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (y_{i,t} \log \nu_{i,t})) - (y_{i,\bullet} + r) \log(\nu_{i,\bullet} + r) \right] \\
&\quad + \sum_{i=1}^{m} \left[\sum_{t=1}^{n_i} (- \log y_{i,t}!) + \log \Gamma(y_{i,\bullet} + r) \right] + mr \log r - m \log \Gamma(r)
\end{aligned}
$$

Note that only purple part is relevant for the estimation of $\boldsymbol{\beta}$.

Poisson-gamma GLMM full loglikelihood - in R

```
"negll.MVNB" <- function(parm) {

## declarations of variables
p       <- ncol(x);
beta    <- parm[1:p]
r       <- parm[p+1]
nu      <- exp(x %*% beta+log(offset));
dat     <- cbind(id,nu,y);
y_dot   <- aggregate(y~id,dat,sum)[,2]
nu_dot  <- aggregate(nu~id,dat,sum)[,2]
m       <- length(unique(id))
```

Poisson-gamma GLMM full loglikelihood - in R

```
## full loglikelihood
temp1 = sum(y*log(nu))
                    - sum( (y_dot+r)* log(nu_dot+r) )
temp2 = -sum(lgamma(y+1)) + sum(lgamma(y_dot+r))
                    + m*r*log(r)-m*lgamma(r)
result = -temp1-temp2
return(result)
}
```

Poisson-gamma GLMM full loglikelihood - in R

```
## optimization in R
init.est <- as.vector(c(init.beta, init.r))
# init.est could be borrowed from marginal models

fit.MVNB <- optim(init.est, negll.MVNB,
                  hessian=TRUE)

beta.hat <- fit.MVNB$par[1:p]
r.hat <- fit.MVNB$par[p+1]
# r.hat should be positive!
parm.stderr <- sqrt(diag(solve(fit.MVNB$hessian)))
# parm.stderr should be positive!
loglik.MVNB <- -fit.MVNB$value
```

Poisson-gamma GLMM - simulation study

To confirm that the proposed Poisson-gamma GLMM properly works, we generate $\{y_{i,t}\}_{i=1,\ldots,50000,\, t=1,\ldots,6}$ as follows:

$$y_{i,t}|\Theta_i \overset{indep.}{\sim} \mathcal{P}(\Theta_i \nu_{i,t}),\ \Theta_i \overset{i.i.d.}{\sim} \mathcal{G}(r, 1/r),$$

$$x_{1,i,t} \sim \mathcal{N}(0,1),\ x_{2,i,t} \sim \mathcal{BN}(0.6),$$

and

$$\nu_{i,t} = m_{i,t} \cdot e^{\mathbf{x}'_{i,t}\boldsymbol{\beta}} = \exp(\beta_0 + \beta_1 x_{1,i,t} + \beta_2 x_{2,i,t} + \log m_{i,t}),$$
$$\beta_0 = -2.5,\ \beta_1 = 1.5,\ \beta_2 = 0.20,\ r = 5.$$

We use the data points from $t = 1, \ldots, 5$ for the estimation purpose, and those from $t = 6$ for the out-of-sample validation purpose, respectively.

Poisson-gamma GLMM - estimation results

	Poisson		MVNB	
	Estimate	Std. Error	Estimate	Std. Error
β_0	-2.5063	0.0110	-2.5053	0.0113
β_1	1.4963	0.0096	1.4959	0.0100
β_2	0.2115	0.0120	0.2104	0.0123
r	-	-	4.7422	0.2686
Loglikelihood	-85028.36		-84811.28	
AIC	170062.71		169630.56	
BIC	170094.00		169647.42	

Poisson-gamma GLMM - estimation results

We observe that

- Interestingly, both the Poisson GLM and Poisson-gamma GLMM (MVNB) produce reliable results in the estimation of $\boldsymbol{\beta}$.
- In fact, as long as the marginal mean is well-specified (here $\mathbb{E}[y_{i,t}] = \nu_{i,t}$ for both models), then $\hat{\boldsymbol{\beta}}$ is consistent (very close to the true $\boldsymbol{\beta}$ in case the sample size is very large). Such idea was extensively explored in the statistics literature, which is known as GEE (Generalized estimating equation, not this!).
- However, it is confirmed that the Poisson-gamma GLMM (MVNB) improves the goodness-of-fit significantly.

We will later analyze how these two models are working in terms of the out-of-sample validation (prediction).

Poisson-gamma GLMM - prediction

Recall that we can obtain the joint probability function of $(y_{i,1}, \ldots, y_{i,n_i})$ as follows:

$$h(y_{i,1}, \ldots, y_{i,n_i}) = \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \frac{\Gamma(y_{i,\bullet} + r)}{(\nu_{i,\bullet} + r)^{y_{i,\bullet}+r}}.$$

Can we compute the predictive probability function of y_{i,n_i+1} given $(y_{i,1}, \ldots, y_{i,n_i})$?

Poisson-gamma GLMM - prediction

Recall that we can obtain the joint probability function of $(y_{i,1}, \ldots, y_{i,n_i})$ as follows:

$$h(y_{i,1}, \ldots, y_{i,n_i}) = \left[\prod_{t=1}^{n_i} \frac{(\nu_{i,t})^{y_{i,t}}}{y_{i,t}!} \right] \frac{r^r}{\Gamma(r)} \times \frac{\Gamma(y_{i,\bullet} + r)}{(\nu_{i,\bullet} + r)^{y_{i,\bullet}+r}}.$$

Can we compute the predictive probability function of y_{i,n_i+1} given $(y_{i,1}, \ldots, y_{i,n_i})$? YES!

Poisson-gamma GLMM - prediction

By using the conditioning argument, we have

$$f(y_{i,n_i+1}|y_{i,1},\ldots,y_{i,n_i}) = \frac{h(y_{i,1},\ldots,y_{i,n_i},y_{i,n_i+1})}{h(y_{i,1},\ldots,y_{i,n_i})}$$

$$= \left[\frac{(\nu_{i,n_i+1})^{y_{i,n_i+1}}}{y_{i,n_i+1}!}\right] \times \frac{(\nu_{i,\bullet}+r)^{y_{i,\bullet}+r}}{\Gamma(y_{i,\bullet}+r)} \times \frac{\Gamma(y_{i,\bullet}+y_{i,n_i+1}+r)}{(\nu_{i,\bullet}+\nu_{i,n_i+1}+r)^{y_{i,\bullet}+y_{i,n_i+1}+r}}$$

$$= \frac{\Gamma(y_{i,\bullet}+y_{i,n_i+1}+r)}{y_{i,n_i+1}!\Gamma(y_{i,\bullet}+r)} \times$$

$$\left(\frac{\nu_{i,\bullet}+r}{\nu_{i,\bullet}+\nu_{i,n_i+1}+r}\right)^{y_{i,\bullet}+r} \left(\frac{\nu_{i,n_i+1}}{\nu_{i,\bullet}+\nu_{i,n_i+1}+r}\right)^{y_{i,n_i+1}}$$

$$= \frac{\Gamma(y_{i,\bullet}+y_{i,n_i+1}+r)}{y_{i,n_i+1}!\Gamma(y_{i,\bullet}+r)} \times$$

$$\left(\frac{1}{1+\nu_{i,n_i+1}/(\nu_{i,\bullet}+r)}\right)^{y_{i,\bullet}+r} \left(\frac{\nu_{i,n_i+1}/(\nu_{i,\bullet}+r)}{1+\nu_{i,n_i+1}/(\nu_{i,\bullet}+r)}\right)^{y_{i,n_i+1}}$$

Poisson-gamma GLMM - prediction

Therefore, we can confirm that

$$y_{i,n_i+1} \,|\, y_{i,1}, \ldots, y_{i,n_i} \sim \mathcal{NB}\left(y_{i,\bullet} + r,\ \frac{\nu_{i,n_i+1}}{\nu_{i,\bullet} + r}\right),$$

so that

$$\mathbb{E}[y_{i,n_i+1} | y_{i,1}, \ldots, y_{i,n_i}] = (y_{i,\bullet} + r) \cdot \frac{\nu_{i,n_i+1}}{\nu_{i,\bullet} + r} = \frac{y_{i,\bullet} + r}{\nu_{i,\bullet} + r} \cdot \nu_{i,n_i+1}.$$

Poisson-gamma GLMM - prediction

We may interpret the above predictive premium as follows:

$$\mathbb{E}[y_{i,n_i+1}|y_{i,1},\ldots,y_{i,n_i}] = \frac{y_{i,\bullet}+r}{\nu_{i,\bullet}+r}\cdot\nu_{i,n_i+1}$$

Poisson-gamma GLMM - prediction

We may interpret the above predictive premium as follows:

$$\begin{aligned}\mathbb{E}[y_{i,n_i+1}|y_{i,1},\ldots,y_{i,n_i}] &= \frac{y_{i,\bullet}+r}{\nu_{i,\bullet}+r}\cdot\nu_{i,n_i+1}\\ &= \mathbb{E}[\Theta_i|y_{i,1},\ldots,y_{i,n_i}]\cdot\mathbb{E}[y_{i,n_i+1}]\end{aligned}$$

Poisson-gamma GLMM - prediction

We may interpret the above predictive premium as follows:

$$\begin{aligned}\mathbb{E}[y_{i,n_i+1}|y_{i,1},\ldots,y_{i,n_i}] &= \frac{y_{i,\bullet}+r}{\nu_{i,\bullet}+r}\cdot\nu_{i,n_i+1}\\ &= \mathbb{E}[\Theta_i|y_{i,1},\ldots,y_{i,n_i}]\cdot\mathbb{E}[y_{i,n_i+1}]\\ &= \frac{\text{(actual past \# of claims)}}{\text{(expected past \# of claims)}}\times\text{(expected current \# of claims)}.\end{aligned}$$

Therefore, the predictive premium from a Poisson-gamma GLMM also shares the spirit of **credibility** premium!

Poisson GLM - prediction

In the case of a Poisson GLM, we assume that all the responses $\{y_{i,t}\}$ are independent so that $h(y_{i,1}, \ldots, y_{i,n_i}) = \prod_{t=1}^{n_i} f(y_{i,t})$, and

$$f(y_{i,n_i+1} | y_{i,1}, \ldots, y_{i,n_i}) = \frac{h(y_{i,1}, \ldots, y_{i,n_i}, y_{i,n_i+1})}{h(y_{i,1}, \ldots, y_{i,n_i})}$$

Poisson GLM - prediction

In the case of a Poisson GLM, we assume that all the responses $\{y_{i,t}\}$ are independent so that $h(y_{i,1}, \ldots, y_{i,n_i}) = \prod_{t=1}^{n_i} f(y_{i,t})$, and

$$f(y_{i,n_i+1}|y_{i,1}, \ldots, y_{i,n_i}) = \frac{h(y_{i,1}, \ldots, y_{i,n_i}, y_{i,n_i+1})}{h(y_{i,1}, \ldots, y_{i,n_i})}$$

$$= \frac{\prod_{t=1}^{n_i+1} f(y_{i,t})}{\prod_{t=1}^{n_i} f(y_{i,t})} = f(y_{i,n_i+1}),$$

and subsequently

$$\mathbb{E}[y_{i,n_i+1}|y_{i,1}, \ldots, y_{i,n_i}] = \mathbb{E}[y_{i,n_i+1}] = \nu_{i,n_i+1}.$$

It implies that in the case of Poisson GLM, past observations **DO NOT** affect the future premium due to the (implicitly assumed) independence among the observed claims.

Out-of-sample validation for claim frequency

It is natural to use the following measures, root-mean-squared error (RMSE) and Poisson deviance (PDEV) to assess the prediction performance:

$$\text{RMSE}: \sqrt{\frac{1}{m}\sum_{i=1}^{M}(y_{i,T_i+1}-\hat{y}_{i,T_i+1})^2},$$

$$\text{PDEV}: \frac{2}{m}\sum_{i=1}^{m}\left(y_{i,T_i+1}\log\frac{y_{i,T_i+1}}{\hat{y}_{i,T_i+1}}-y_{i,T_i+1}+\hat{y}_{i,T_i+1}\right).$$

We prefer a model with lower RMSE/PDEV.

Out-of-sample validation for claim frequency

	Poisson	MVNB
RMSE	0.3565	0.3557
PDEV	0.4595	0.4564

It is shown that usage of Poisson-gamma GLMM slightly improves the prediction performance.

Part 2: Credibility premium of S with the static random effects

P&C Insurance Claim Data Structure

- Let N be the number of accidents and $Y_1, \cdots, Y_N$ be the size of claim per accident.
- For ratemaking in P&C, we have to predict the cost of claims $S_{it} = \sum_{k=1}^{N_{it}} Y_{itk}$.
- Policyholder i is followed over time $t = 1, \ldots, T_i$ years.
- Unit of analysis "it" – an insured driver i over time t (year)
- For each "it", could have several claims, $k = 0, 1, \ldots, N_{it}$
- Have available information on: number of claims N_{it}, amount of claim Y_{itk}, exposure e_{it} and covariates (explanatory variables) x_{it}
  - covariates often include age, gender, vehicle type, building type, building location, driving history and so forth

The Two-Part Frequency-Severity Model

- We will model the pair (N_{it}, C_{it}) where

$$C_{it} = \begin{cases} \frac{1}{N_{it}} \sum_{k=1}^{N_{it}} Y_{itk}, & N_{it} > 0 \\ \textit{Undefined}, & N_{it} = 0 \end{cases}$$

  is the observed average claim size and $S_{it} = \sum_{k=1}^{N_{it}} Y_{itk}$ is the observed aggregate claim size.

- The joint density of the number of claims and the average claim size can be decomposed as

$$\begin{aligned} f(N, C|\mathbf{x}) &= f(N|\mathbf{x}) \times f(C|N, \mathbf{x}) \\ \text{joint} &= \text{frequency} \times \text{average severity}. \end{aligned}$$

Ratemaking with Predictive Analytics

[Observed] $\{x_{it}, N_{it}, C_{it}, e_{it} | i = 1, 2, \ldots, I, t = 1, 2, \ldots, T_i\}$

$\Downarrow$ Calibrated Model $\Downarrow$

[Predicted] N_{i,T_i+1}, C_{i,T_i+1}, and S_{i,T_i+1}

Ratemaking with Predictive Analytics

$$\text{[Observed]} \quad \{x_{it}, N_{it}, C_{it}, e_{it} | i = 1, 2, \ldots, I, t = 1, 2, \ldots, T_i\}$$

$$\Downarrow \text{ Calibrated Model } \Downarrow$$

$$\text{[Predicted]} \quad N_{i,T_i+1},\ C_{i,T_i+1},\ \text{and}\ S_{i,T_i+1}$$

From now on, we drop the subscript i for succint expressions.

Why Random Effects?

The following hypothetical example shows us that we might capture the unobserved heterogeneity in risk by observing the residual after controlling for the effects of observed covariates, which can be explained in terms of random effects for policyholders A and B.

Year	Gender	Age	Vehicle Size	Policyholder A		Policyholder B	
				# of Claim	Amt of Claim	# of Claim	Amt of Claim
2015	M	45	Medium	0	0	1	500
2016	M	46	Medium	0	0	2	4000
2017	M	47	Large	1	200	1	8000

Note that random effects model has a good connection with credibility theory as well.

Frequency Model Specifications

In actuarial practice, Poisson distribution has been used for the calibration of the number of claim with the presence of covariates as follows:

$$N_t|\mathbf{x}_t, e_t \overset{indep}{\sim} \mathcal{P}(\nu_t) \quad \text{where} \quad \nu_t = e_t \exp(\mathbf{x}_t\alpha). \tag{3}$$

Due to the longitudinal property of usual claim dataset, we may consider the following extension by incorporating random effects as follows:

$$N_t|\theta^N \overset{indep}{\sim} \mathcal{P}(\nu_t\theta^N) \quad \text{where} \quad \nu_t = e_t \exp(\mathbf{x}_t\alpha),\ \theta^N \sim \pi_N(\theta). \tag{4}$$

Frequency Model Specifications

Let us consider the following frequency models:

- For the count of claims (frequency)
  (1) Simple Poisson GLM: $N_t \sim \text{Pois}(e^{\mathbf{x}_t\alpha})$ so that $\mathbb{E}[N_t|\mathbf{x}_t] = e^{\mathbf{x}_t\alpha}$.
  (2) Poisson/Gamma Random Effect Model (= Multivariate NB Model)

Poisson/Gamma Random Effect Model

Data likelihood: $N_t|\theta^N \overset{indep}{\sim} \mathcal{P}(\nu_t\theta^N)$
Prior density: $\pi_N(\theta) \propto \theta^{r-1}e^{-\theta r}$ so that $\theta^N \sim \mathcal{G}(r,r)$ and $\mathbb{E}\left[\theta^N\right] = 1, \mathrm{V}ar\left(\theta^N\right) = \frac{1}{r}$.
Posterior density:
$\theta^N|n_1, n_2, \ldots, n_T \sim \mathcal{G}(\sum_{t=1}^T n_t + r, \sum_{t=1}^T \nu_t + r)$
Predictive distribution:

$$N_{T+1}|n_1, n_2, \ldots, n_T \sim \mathcal{NB}\left(\sum_{t=1}^T n_t + r, \frac{\nu_{T+1}}{\sum_{t=1}^T \nu_t + r}\right) \tag{5}$$

so that $\mathbb{E}\left[N_{T+1}|n_1, n_2, \ldots, n_T\right] = \frac{\sum_{t=1}^T n_t + r}{\sum_{t=1}^T \nu_t + r}\nu_{T+1}$.

Implication of the Frequency Credibility Premium

- We can see that

$$\mathbb{E}\left[N_{T+1}|\mathcal{F}_T\right] = \frac{r + \sum_{t=1}^{T} n_t}{r + \sum_{t=1}^{T} \exp(\mathbf{x}_t \alpha)} \times \exp(\mathbf{x}_{T+1}\alpha)$$

$$= \frac{(\text{actual past } \# \text{ of claims})}{(\text{expected past } \# \text{ of claims})} \times (\text{expected current } \# \text{ of claims}).$$

- In this sense, use of random effects helps us to penalize/reward a policyholder based on his/her driving experience.
- What about $\mathbb{E}\left[S_{T+1}|\mathcal{F}_T\right]$?

Static Credibility Premium for S when $N \perp C$

Frangos and Vrontos (2001) derived the following formula for (static) credibility premium of S assuming $N \perp C$.

Theorem

Suppose $(N_1, N_2, \ldots, N_t)$ follows MVNB distribution as defined in (5). Moreover, let us assume that $\mathbb{E}[C_t|n_t] = \theta^C e^{\mathbf{x}_t\beta} = \theta^C \tilde{\mu}_t$. Then credibility premium of $S_{i,T+1}$ given $n_1, \ldots, n_T, c_1, \ldots, c_T$ is given as follows:

$$\begin{aligned}\mathbb{E}[S_{T+1}|\mathcal{F}_T] &= \mathbb{E}[N_{T+1}|n_1, \ldots, n_T] \times \mathbb{E}[C_{T+1}|\mathcal{F}_T] \\ &= \textit{Cred. Freq. mean } \times \textit{Cred. Sev. mean}\end{aligned}$$

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What if $N \not\perp C$?

Cross-sectional Dependence between N and C

Inspired by Garrido et al. (2016), we assume a specific form of cross-sectional dependence between N_t and C_t as follows:

$$
\begin{aligned}
&N_t|\theta^N \overset{indep}{\sim} \mathcal{P}(\nu_t\theta^N) \quad \text{where} \quad \nu_t = e_t \exp(\mathbf{x}_t\alpha),\ \theta^N \sim \mathcal{G}(r,r).\\
&C_t|\theta^C, n_t \overset{indep}{\sim} \mathcal{G}\left(\frac{n_t}{\psi}, \frac{n_t}{\psi\theta^C\mu_t}\right) \quad \text{where} \quad \mu_t = \tilde{\mu}_t e^{\gamma n_t} = \exp(\mathbf{x}_t\beta + \gamma n_t),\\
&\theta^C \sim \mathcal{IG}(k+1,k).
\end{aligned}
\tag{6}
$$

Since $\mu_t = \exp(\mathbf{x}_t\beta + \gamma n_t)$,
$\mathbb{E}\left[S_t|\theta^N,\theta^C\right] \neq \mathbb{E}\left[N_t|\theta^N\right]\mathbb{E}\left[C_t|\theta^C,n_t\right]$.

Severity Model Specifications

Let us consider the following severity models:

- For the average size of claims (severity)
  (1) Simple Gamma GLM: $C_t|n_t \sim \mathcal{G}(\frac{n_t}{\psi}, \frac{n_t}{\psi\mu_t})$ so that $\mathbb{E}[C_t|n_t, \mathbf{x}_t] = \mu_t = \tilde{\mu}_t e^{\gamma n_t}$ and $\frac{\mathrm{Var}(C_t|n_t,\mathbf{x}_t)}{\mathbb{E}[C_t|n_t,\mathbf{x}_t]^2} = \frac{\psi}{n_t}$
  (2) Gamma/I-gamma Random Effect Model (= Multivariate GP Model)

Gamma/I-gamma Random Effects Model

Data likelihood: $C_t|\theta^C \overset{indep}{\sim} \mathcal{G}\left(\frac{n_t}{\psi}, \frac{n_t}{\psi\theta^C\mu_t}\right)$

Prior density: $\pi_C(\theta) \propto \theta^{-k-2}e^{-k/\theta}$ so that $\theta^C \sim \mathcal{IG}(k+1, k)$ and $\mathbb{E}\left[\theta^C\right] = 1, \mathrm{V}ar\left(\theta^C\right) = \frac{1}{k-1}$.

Posterior density:
$\theta^C|\mathcal{F}_T \sim \mathcal{IG}(k + \sum_{t=1}^{T} n_t/\psi + 1, k + \sum_{t=1}^{T} \frac{s_t}{\psi\mu_t})$.

Note that $s_t = n_t \cdot c_t$ by definition.

Gamma/I-gamma Random Effects Model

Predictive distribution:

$$C_{T+1}|\mathcal{F}_T, n_{T+1} \sim \mathcal{GP}\left(k + \sum_{t=1}^{T} n_t/\psi + 1, \left[k + \sum_{t=1}^{T} \frac{s_t}{\psi\mu_t}\right] \frac{\psi\mu_{T+1}}{n_{T+1}}, \frac{n_{T+1}}{\psi}\right),$$

with predictive mean

$$\mathbb{E}\left[C_{T+1}|n_{T+1}, \mathcal{F}_T\right] = \frac{\left[k\psi + \sum_{t=1}^{T} s_t/\mu_t\right]}{k\psi + \sum_{t=1}^{T} n_t}\mu_{T+1}.$$

GP and GB2 Distributions

- $Y \sim \mathcal{GP}(a, \xi, \tau)$

$$\implies \quad f(y|a, \xi, \tau) = \frac{\Gamma(a+\tau)}{\Gamma(a)\Gamma(\tau)} \frac{\xi^a y^{\tau-1}}{(y+\xi)^{a+\tau}}$$

- $Y \sim \mathcal{GB}2(a, \xi, \tau, p)$

$$\implies \quad f(y|a, \xi, \tau, p) = \frac{\Gamma(a+\tau)}{\Gamma(a)\Gamma(\tau)} |p| \frac{\xi^{ap} y^{\tau p-1}}{(y^p+\xi^p)^{a+\tau}}$$

- Note that GP distribution is a special case of GB2 distribution where $p = 1$. (Loss Models, Klugman et al. 2012)

Static Credibility Premium for S when $N \not\perp C$

Theorem

Suppose $(N_1, N_2, \ldots, N_t)$ *follows MVNB distribution as defined in (5). Moreover, let us assume that* $\mathbb{E}[C_t|n_t] = \theta^C e^{\mathbf{x}_t\beta}\, e^{n_t\gamma} = \theta^C \tilde{\mu}_t\, e^{n_t\gamma}$. *Then, the credibility premium of* S_{T+1} *is given as follows:*

$$\mathbb{E}[S_{T+1}|\mathcal{F}_T] = \mathbb{E}\left[\tilde{\mu}_{T+1}\theta^C|\mathcal{F}_T\right] \times \mathbb{E}\left[\nu_{T+1}\theta^N|\underline{n_{t-1}}\right] \times e^{\gamma}\left[1 - \left(\frac{\nu_{T+1}}{\sum_{t=1}^{T}\nu_t + r}\right)(e^{\gamma}-1)\right]^{-\sum_{t=1}^{T} n_t - r - 1}. \tag{7}$$

If $\gamma = 0$, *then (7) is reduced as in Frangos and Vrontos (2001):*

$$\mathbb{E}[S_{T+1}|\mathcal{F}_T] = \mathbb{E}\left[\tilde{\mu}_{T+1}\theta^C|\mathcal{F}_T\right] \times \mathbb{E}\left[\nu_{T+1}\theta^N|\underline{n_{t-1}}\right].$$

Static Credibility Premium for S when $N \not\perp C$

- According to the theorem, it can be observed that the **static** credibility premium of S_{T+1} can be expressed as the product of
  - **static** credibility premium of N_{T+1},
  - **static** credibility premium of C_{T+1}, and
  - adjustment factor which accounts for the dependence between frequency and the average severity.
- Under the MVNB distribution defined in (5), we have the following:

$$\mathbb{E}\left[\nu_{T+1}\theta^{N}|\underline{n_{t-1}}\right] = \frac{r+\sum_{t=1}^{T} n_t}{r+\sum_{t=1}^{T}\nu_t}\nu_{T+1}.$$

- For convenience, let us denote $D_N(\gamma) = e^{\gamma}\left[1-\left(\frac{\nu_{T+1}}{\sum_{t=1}^{T}\nu_t+r}\right)(e^{\gamma}-1)\right]^{-\sum_{t=1}^{T} n_t-r-1}$ so that $D_N(0) = 1$.

Credibility Premium for the Compound Sum when $N \not\perp C$

Theorem

For $D_N(\gamma)$ in (7), the following are true:

(i) *$D_N(\gamma)$ is well-defined if and only if $\gamma \leq \log\left(1 + \frac{r+\sum_{t=1}^{T}\nu_t}{\nu_{T+1}}\right)$.*

(ii) *$D_N(\gamma)$ is a strictly increasing function of γ.*

(iii)

$$D_N(\gamma) = e^{\gamma}\left[1 - \left(\frac{\nu_{T+1}}{r+\sum_{t=1}^{T}\nu_t}\right)(e^{\gamma}-1)\right]^{-r-\sum_{t=1}^{T}n_t-1} \begin{cases} = 1 & \text{if} \quad \gamma = 0 \\ > 1 & \text{if} \quad \gamma > 0 \\ < 1 & \text{if} \quad \gamma < 0 \end{cases}$$

In this regard, $D_N(\gamma)$ can be interpreted as an informative measure of dependence which considers not only the sign of dependence but also the magnitude of dependence.

Part 3: Credibility premium of S with the dynamic random effects

Static Frequency Credibility Premium

- Recall that in Part 1, the frequency credibility premium is given as follows:

$$\mathbb{E}\left[N_{T+1}|\mathcal{F}_T\right] = \frac{r + \sum_{t=1}^{T} n_t}{r + \sum_{t=1}^{T} \exp(\mathbf{x}_t \alpha)} \times \exp(\mathbf{x}_{T+1}\alpha)$$

$$= \frac{(\text{actual past } \# \text{ of claims})}{(\text{expected past } \# \text{ of claims})} \times (\text{expected current } \# \text{ of claims}).$$

Static Frequency Credibility Premium

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$$= \frac{\text{(actual past \# of claims)}}{\text{(expected past \# of claims)}} \times \text{(expected current \# of claims)}.$$

- In this sense, using **static** random effects is somewhat equivalent to use aggregated actual/expected # of claims for adjustments.

Static Frequency Credibility Premium

- Recall that in Part 1, the frequency credibility premium is given as follows:

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$$= \frac{(\text{actual past } \# \text{ of claims})}{(\text{expected past } \# \text{ of claims})} \times (\text{expected current } \# \text{ of claims}).$$

- In this sense, using **static** random effects is somewhat equivalent to use aggregated actual/expected # of claims for adjustments.
- However, it means that there is no distinction on the age of claims. (i.e., $n_1 = 1$ or $n_5 = 1$ ends up with the same impacts in the credibility premium.)

Disclaimer: STATIC Random Effects Models

- In the formulation, we assumed that the random effect θ_i is random but STATIC. In other words, it is hard to capture possible evolution on the unobserved heterogeneity in risks over time with such model.

Disclaimer: STATIC Random Effects Models

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- In the formulation, we assumed that the random effect θ_i is random but STATIC. In other words, it is hard to capture possible evolution on the unobserved heterogeneity in risks over time with such model.
- This assumption leads to the predictive premium formula which assigns the same weight for every n_t, $t = 1, \ldots, T$.
- One can suggest DYNAMIC random effects compound risk models (CRMs) to capture possible evolution on the unobserved heterogeneity, which follows in the later part of this presentation.

Key Idea of Static Random Effects CRMs

$$
\begin{array}{ccccccccc}
\cdots & = & \theta^N & = & \theta^N & = & \theta^N & = \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & \\
\cdots & & N_{t-1} & & N_t & & N_{t+1} & \\
 & & \Downarrow & & \Downarrow & & \Downarrow & \\
\cdots & & C_{t-1} & & C_t & & C_{t+1} & \\
\uparrow & & \uparrow & & \uparrow & & \uparrow & \\
\cdots & = & \theta^C & = & \theta^C & = & \theta^C & =
\end{array}
$$

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 & & \Downarrow & & \Downarrow & & \Downarrow & \\
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- θ^N determines the conditional distribution of N_{t-1} (respectively, C).

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 & & \Downarrow & & \Downarrow & & \Downarrow & \\
\cdots & & C_{t-1} & & C_t & & C_{t+1} & \\
\uparrow & & \uparrow & & \uparrow & & \uparrow & \\
\cdots & = & \theta^C & = & \theta^C & = & \theta^C & =
\end{array}
$$

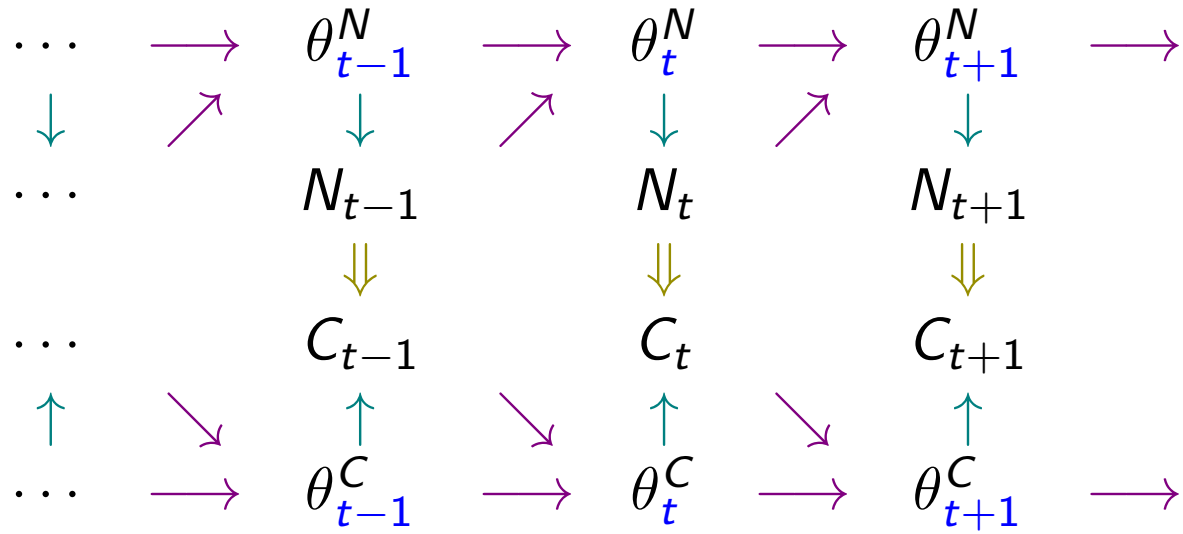
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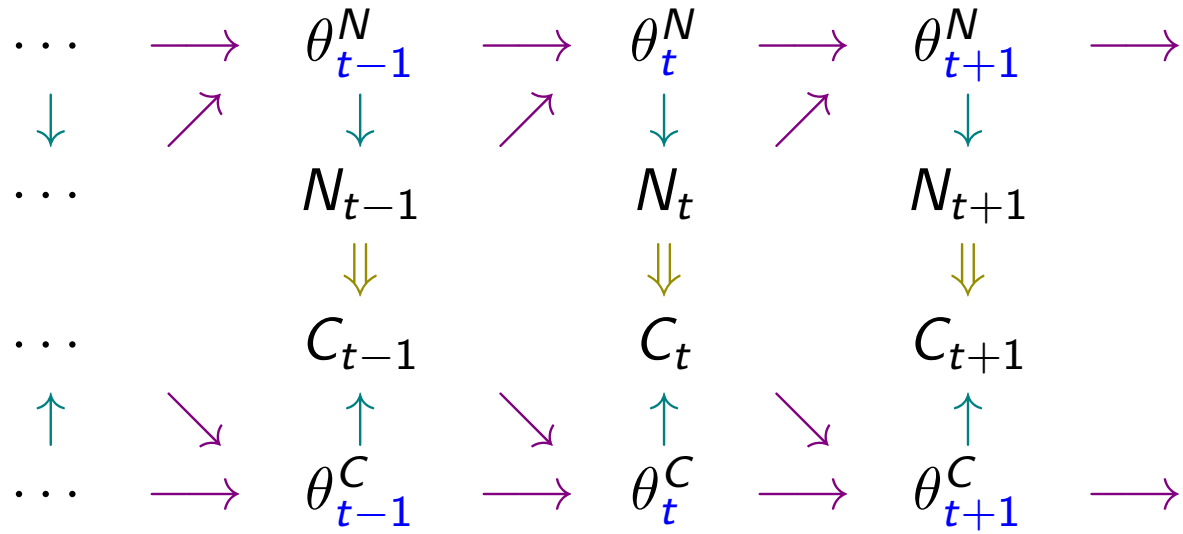
$$
\begin{array}{ccccccccc}
\cdots & = & \theta^N & = & \theta^N & = & \theta^N & = \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & \\
\cdots & & N_{t-1} & & N_t & & N_{t+1} & \\
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\uparrow & & \uparrow & & \uparrow & & \uparrow & \\
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\end{array}
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- N_t is assumed to have cross-sectional dependence with C_t.

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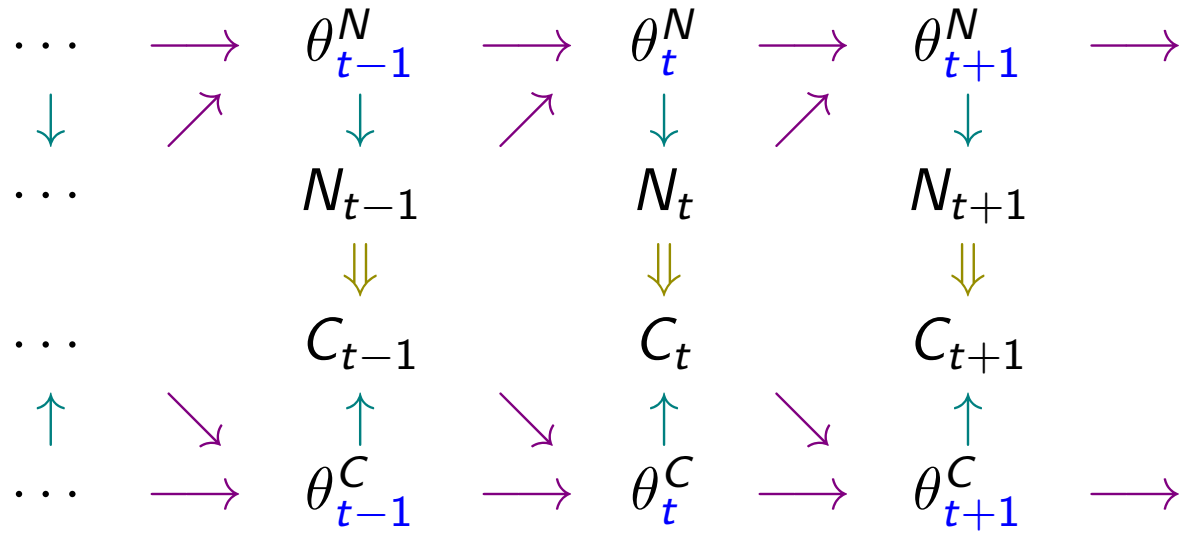


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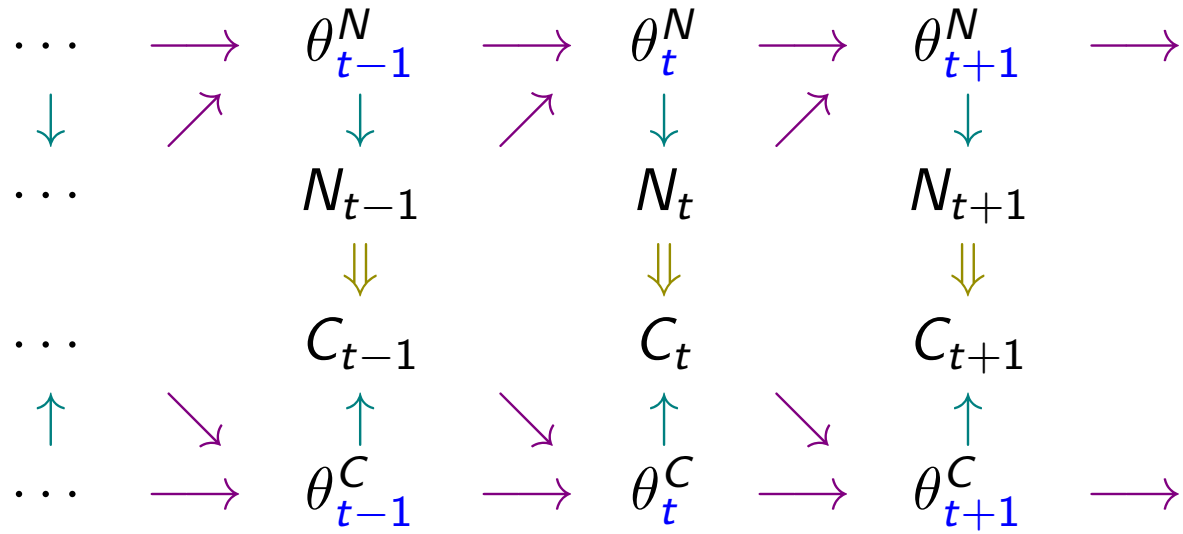
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- θ^N_{t-1} and N_{t-1} affect the filtering distribution of θ^N_t (respectively, C).

Key Idea of Dynamic Random Effects CRMs



- θ^N_{t-1} determines the conditional distribution of N_{t-1} (respectively, C).
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Building Blocks for Dynamic Random Effects CRMs

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- Innovation of the dynamic (gamma) random effects (Lukacs, 1995),
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- Cross-sectional dependence between the frequency and severity (Garrido et al., 2016; Jeong and Valdez, 2020).

Innovation for the Dynamic Random Effects

Lemma (Lukacs, 1955)

Given two independent random variables $\theta_t \sim \mathcal{G}(u, v)$ *and* $Z_t \sim \mathcal{B}(wu, (1-w)u)$*, where* $u > 0, v > 0, w \in (0, 1)$ *are constants, we have*

$$\theta_{t+1} := \frac{Z_t \theta_t}{w} \sim \mathcal{G}(wu, wv).$$

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- Z_t: source of (random) innovation at time t for the random effect.

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- θ_t: the dynamic random effect at time t,
- Z_t: source of (random) innovation at time t for the random effect.
- It is easy to check that $\mathrm{Cor}(\theta_t, \theta_{t+1}) = w$.

Indeed, if $w = 1$, then Z_t is almost surely 1 so that $\theta_{t+1} = \theta_t$, which is equivalent to the **static** random effects case.

Frequency Dynamic Random Effects (HF, 1989)

The joint dynamics of the count (i.e. frequency) N_t and the random effect θ_t^N is as follows:

i. (Updating rule of θ_t^N) Assume that at time $t > 1$, the filtering distribution of θ_{t-1}^N given the past observations[1] $\underline{n_{t-1}} := (n_1, \cdots, n_{t-1})$ is given by

$$\theta_{t-1}^N | \underline{n_{t-1}} \sim \mathcal{G}\left(r_{t-1}, r'_{t-1}\right),$$

where $r_1 = r'_1 = r$ and r_{t-1}, r'_{t-1} are positive, deterministic functions of past counts and exogenous covariates observed up to time $t-1$.

[1] We similarly define

$$\underline{A_t} := (A_1, \cdots, A_t), \quad \underline{\theta_t^N} := (\theta_1^N, \cdots, \theta_t^N) \quad \text{and} \quad \underline{\nu_t} := (\nu_1, \cdots, \nu_t).$$

Frequency Dynamic Random Effects (HF, 1989)

Then the new state variable θ_t^N is defined by

$$\theta_t^N := \frac{\theta_{t-1}^N A_t}{p},$$

where $p \in (0,1)$ is a predetermined constant and A_t has the following predictive distribution:

$$A_t \,|\, \underline{A_{t-1}}, \underline{n_{t-1}}, \underline{\theta_{t-1}^N} \sim \mathcal{B}\left(pr_{t-1}, (1-p)r_{t-1}\right).$$

Note that the conditional distribution of θ_t^N given $\underline{n_{t-1}}$, called the predictive distribution, is given by:

$$\theta_t^N | \underline{n_{t-1}} \sim \mathcal{G}\left(pr_{t-1}, pr'_{t-1}\right). \tag{8}$$

Frequency Dynamic Random Effects (HF, 1989)

ii. (Observation at time t) Given $\underline{n_{t-1}}$ and $\underline{\theta_t^N}$, observation N_t has the following conditional distribution:

$$N_t|\underline{n_{t-1}}, \underline{\theta_t^N} \sim \mathcal{P}\left(\nu_t \theta_t^N\right),$$

where ν_t is a positive, deterministic function of exogenous covariates observed at time t. Thus by the Poisson-gamma conjugacy, the filtering distribution of θ_t^N given $\underline{n_t}$ is given by:

$$\theta_t^N|\underline{n_t} \sim \mathcal{G}\left(r_t, r_t'\right), \tag{9}$$

with

$$r_t := pr_{t-1} + n_t > 0 \quad \textit{and} \quad r_t' := pr_{t-1}' + \nu_t > 0.$$

Frequency Dynamic Random Effects (HF, 1989)

- An immediate consequence is the following relationships concerning the predictive mean and variance of the random effect:

$$\mathbb{E}\left[\theta_t^N \mid \underline{n_{t-1}}\right] = \mathbb{E}\left[\theta_{t-1}^N \mid \underline{n_{t-1}}\right], \mathrm{V}ar\left(\theta_t^N \mid \underline{n_{t-1}}\right) = \frac{1}{p}\mathrm{V}ar\left(\theta_{t-1}^N \mid \underline{n_{t-}}\right.$$

- Thus, updating θ_{t-1}^N to θ_t^N has the effect of preserving the mean (as well as the parametric form of the distribution) when passing from the filtering distribution to the predictive distribution.
- It introduces extra "uncertainty", through the multiplicative "innovation" $\frac{A_t}{p}$, which is of unit mean. This transition has also the effect of inflating the variance by a factor of $1/p$.

Frequency Dynamic Random Effects (HF, 1989)

One can check that $N_{T+1} \,|\, \underline{n_T}$ still follows a negative binomial distribution with the following predictive mean:

$$\mathbb{E}\left[N_{T+1} \,|\, \underline{n_T}\right] = \nu_{T+1}\frac{r_T}{r'_T} = \nu_{T+1}\frac{\sum_{t=1}^{T} p^{T-t} n_t + p^T r}{\sum_{t=1}^{T} p^{T-t}\nu_t + p^T r} = \nu_{T+1}\left[a_0 + \sum_{t=1}^{T} a_t \right. \tag{10}$$

$$\text{with } a_0 := \frac{p^T r}{\sum_{t=1}^{T} p^{T-t}\nu_t + p^T r} \quad \text{and} \quad a_t := \frac{p^{\tau-1-t}\nu_t}{\sum_{t=1}^{T} p^{T-t}\nu_t + p^T r}.$$

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What if $p = 1$?

Severity Dynamic Random Effects (SM, 1986)

The joint dynamics of the average claim amounts (i.e. severity) C_t and the random effect θ_t^C is:

i. (Updating rule of θ_t^C) Assume that at time $t > 1$, the filtering distribution of θ_{t-1}^C given the past observations[2] $\underline{c_{t-1}} := (c_1, \cdots, c_{t-1})$ and $\underline{n_{t-1}}$ is given by

$$\theta_{t-1}^C | \underline{c_{t-1}}, \underline{n_{t-1}} \sim \mathcal{IG}\left(k_{t-1}, k'_{t-1}\right),$$

where $k_1 - 1 = k'_1 = k$ and k_{t-1}, k'_{t-1} are positive, deterministic functions of past counts and exogenous covariates observed up to time $t-1$.

---

[2] We similarly define

$$\underline{B_t} := (B_1, \cdots, B_t), \quad \underline{\theta_t^C} := (\theta_1^C, \cdots, \theta_t^C) \quad \text{and} \quad \underline{\mu_t} := (\mu_1, \cdots, \mu_t).$$

Severity Dynamic Random Effects (SM, 1986)

Then the new state variable θ_t^C is defined by

$$\theta_t^C := \frac{\theta_{t-1}^C q_t^*}{B_t},$$

where $q_t^* = \frac{q(k_{t-1}-2)+1}{k_{t-1}-1} \in (0,1)$ and B_t has the following predictive distribution:

$$B_t \,|\, \underline{B_{t-1}}, \underline{c_{t-1}}, \underline{n_{t-1}}, \underline{\theta_{t-1}^C} \sim \mathcal{B}\left(q_t k_{t-1}, (1-q_t)k_{t-1}\right),$$

with $q_t = \frac{q(k_{t-1}-2)+2}{k_{t-1}} \in (2/k_{t-1}, 1)$.
Note that the conditional distribution of θ_t^C given $\underline{c_{t-1}}$, called the predictive distribution, is given by:

$$\theta_t^C | \underline{c_{t-1}}, \underline{n_{t-1}} \sim \mathcal{IG}\left(q_t k_{t-1}, q_t^* k'_{t-1}\right). \tag{11}$$

Severity Dynamic Random Effects (SM, 1986)

iii. (Observation at time t) Given $\underline{c_{t-1}}$, $\underline{n_t}$, and $\underline{\theta_t^C}$, observation C_t has the following conditional distribution:

$$C_t|\underline{c_{t-1}}, \underline{n_t}, \underline{\theta_t^C} \sim \mathcal{G}\left(\frac{n_t}{\psi}, \frac{n_t}{\psi\mu_t\theta_t^C}\right),$$

where μ_t is a positive, deterministic function of exogenous covariates observed at time t. Thus by the gamma-IG conjugacy, the filtering distribution of θ_t^C given $\underline{c_t}$ and $\underline{n_t}$ is given by:

$$\theta_t^C|\underline{c_t}, \underline{n_t} \sim \mathcal{IG}\left(k_t, k_t'\right), \tag{12}$$

with

$$k_t := q_t k_{t-1} + n_t/\psi > 0 \quad \textit{and} \quad k_t' := q_t^* k_{t-1}' + \frac{s_t}{\psi\mu_t} > 0.$$

Severity Dynamic Random Effects (SM, 1986)

- An immediate consequence is the following relationships concerning the predictive mean and variance of the random effect:

$$\mathbb{E}\left[\theta_t^C \,|\, \mathcal{F}_{t-1}\right] = \mathbb{E}\left[\theta_{t-1}^C \,|\, \mathcal{F}_{t-1}\right], \mathrm{V}ar\left(\theta_t^C \,|\, \mathcal{F}_{t-1}\right) = \frac{1}{q}\mathrm{V}ar\left(\theta_{t-1}^C \,|\, \mathcal{F}_t\right.$$

- Thus, updating θ_{t-1}^C to θ_t^C has the effect of preserving the mean (as well as the parametric form of the distribution) when passing from the filtering distribution to the predictive distribution.
- It introduces extra "uncertainty", through the multiplicative "innovation" $\frac{B_t}{q}$, which is of unit mean. This transition has also the effect of inflating the variance by a factor of $1/q$.

Severity Dynamic Random Effects (SM, 1986)

One can check that $C_{T+1} \,|\, \mathcal{F}_T, n_{T+1}$ still follows a GP distribution with the following predictive mean:

$$\mathbb{E}\left[C_{T+1} \,|\, \mathcal{F}_T, n_{T+1}\right] = \mu_{T+1} \frac{k'_T}{k_T - 1} = \mu_{T+1} \left[b_0 + \sum_{t=1}^{T} b_t \frac{s_t}{\mu_t} \right], \tag{13}$$

$$\text{with } b_0 := \frac{k \prod_{s=1}^{T} q_s^*}{k_T - 1} \quad \text{and} \quad b_t := \frac{\prod_{s=t+1}^{T} q_s^* / \psi}{k_T - 1}.$$

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What if $q = 1$? (Note that $q_s^* = q_s = 1$ for all s if $q = 1$, by definition)

Dynamic Credibility Premium for S when $N \not\perp C$

Theorem

Suppose the predictive means of N_{T+1} and C_{T+1} are given as in (10) and (13), respectively. Then, the (dynamic) credibility premium of S_{T+1} is given as follows:

$$\begin{aligned}\mathbb{E}\left[S_{T+1}|\mathcal{F}_T\right] = \tilde{\mu}_{T+1}\frac{k'_T}{k_T-1} \times \nu_{T+1}\frac{r_T}{r'_T} \\ \times e^{\gamma}\left[1-\left(\frac{\nu_{T+1}}{r'_T}\right)\left(e^{\gamma}-1\right)\right]^{-r_T-1}.\end{aligned} \tag{14}$$

If $\gamma = 0$, then (14) is reduced as follows:

$$\mathbb{E}\left[S_{T+1}|\mathcal{F}_T\right] = \tilde{\mu}_{T+1}\frac{k'_T}{k_T-1} \times \nu_{T+1}\frac{r_T}{r'_T}.$$

Dynamic Credibility Premium for S when $N \not\perp C$

- According to the theorem, it can be observed that the **dynamic** credibility premium of S_{T+1} can be expressed as the product of
  - **dynamic** credibility premium of N_{T+1},
  - **dynamic** credibility premium of C_{T+1}, and
  - adjustment factor which accounts for the dependence between frequency and the average severity.

Data

We use Wisconsin Local Government Property Insurance Fund (LGPIF) dataset. While claim information on multiple lines of business is available, we focus on IM (inland marine) claims and their associated policy characteristics from the selected dataset.

Categorical variables	Description		Proportions	
TypeCity	Indicator for city entity:	Y=1	14.51 %	
TypeCounty	Indicator for county entity:	Y=1	5.92 %	
TypeMisc	Indicator for miscellaneous entity:	Y=1	10.78 %	
TypeSchool	Indicator for school entity:	Y=1	29.1 %	
TypeTown	Indicator for town entity:	Y=1	16.6 %	
TypeVillage	Indicator for village entity:	Y=1	23.09 %	
Continuous variables		Minimum	Mean	Maximum
CoverageIM	Log coverage amount of IM claim	0.00	0.87	46.75
lnDeductIM	Log deductible amount of IM claim	0.00	5.34	9.21

Benchmark models

For comparison, we introduce three benchmark models, which are special or limiting cases of the proposed dynamic random effects model:

- The naive GLMs: $p = q = 1$, $r = k = \infty$, and $\eta = 0$,
- Frequency-severity dependent GLMs (Garrido et al., 2016): $p = q = 1$, $r = k = \infty$,
- Static random effects model (Jeong and Valdez, 2020): $p = q = 1$, and
- Dynamic random effects model.

Benchmark models

For estimation of the parameters, we take a two-step approach so that

- the fixed effects parameters are estimated first due to the common mean structure via usual GLMs, and the dependence structure is considered later. In other words, we fit a usual Poisson/Gamma GLMs to estimate α, β, and γ first,
- and the remaining parameters for the dependence structures (p, q, r, k, and ψ) are estimated afterward.

Regression Estimates of the Fixed Effects

	Frequency		Severity			
			Independent		Dependent	
	Estimate	p-value	Estimate	p-value	Estimate	p-value
(Intercept)	-4.2571	0.0000	9.4271	0.0000	10.1312	0.0000
TypeCity	0.9673	0.0000	1.0040	0.0339	1.3007	0.0012
TypeCounty	1.8747	0.0000	1.3164	0.0060	1.3152	0.0009
TypeMisc	-2.7453	0.0068	-1.3007	0.6018	-1.2977	0.5252
TypeSchool	-0.9174	0.0008	-0.2046	0.7617	-0.1191	0.8300
TypeTown	-0.4129	0.1254	0.2571	0.6981	0.3588	0.5089
CoverageIM	0.0687	0.0000	0.0008	0.9712	0.0332	0.0746
lnDeductIM	0.1363	0.0030	-0.1336	0.4481	-0.1744	0.2278
Count					-0.4538	0.0000

Parameter Estimation for the Dependence Structure

- Static random effects model: $\hat{r} = 0.6143$, $\hat{k} = 1.2615$, and $\hat{\psi} = 1.5145$.
- Dynamic random effects model: $\hat{p} = 0.8202$, $\hat{q} = 0.7615$, $\hat{r} = 0.9228$, $\hat{k} = 1.8761$, and $\hat{\psi} = 1.4671$.

Parameter Estimation for the Dependence Structure

- Recall that

$$\theta_1^N \sim \mathcal{G}(r, r), \qquad \mathbb{E}\left[\theta_1^N\right] = 1, \ \mathrm{Var}\left(\theta_1^N\right) = \frac{1}{r},$$
$$\theta_1^C \sim \mathcal{IG}(k+1, k), \qquad \mathbb{E}\left[\theta_1^C\right] = 1, \ \mathrm{Var}\left(\theta_1^C\right) = \frac{1}{k-1}.$$

  Since $r > k-1$ with the estimates from both static and dynamic models, it implies that the variability of θ_1^C, the (a priori) unobserved heterogeneity for the severity part, is greater than that of θ_1^N, the (a priori) unobserved heterogeneity for the frequency part.

- Secondly, p is estimated to be greater than q in the dynamic model, which implies the decaying impact according to the ages of claims in the determination of the credibility factors would be much stronger in the severity part.

Out-of-sample Validation

	Naive	Garrido	Static	Dynamic
RMSE	9272.96	6433.21	6613.20	6608.58
MAE	1345.41	1241.95	1190.02	1166.10
Portfolio Mean	979.94	905.95	884.05	852.36

* Note that the actual portfolio mean is 645.25.

Implication of Static vs. Dynamic Random Effects

Let us consider a hypothetical scenario where

- $\nu_t = 0.2$ and $\mu_t = 15000$ for $t = 1, \ldots, 5$, and
- $r = 1$, $k = 2$, $\gamma = 0$, and $\psi = 1.5$.

Suppose that there was exactly one claim by a policyholder in one of the years 1 through 4 (i.e., $\sum_{t=1}^{4} n_t = 1$). As shown in the following table, we observe decaying effects of past claims in the calculation of credibility factors under the proposed dynamic model, whereas the static random effects model disregards such information.

Implication of Static vs. Dynamic Random Effects

	Dynamic ($p = q = 0.8$)		Static ($p = q = 1$)	
Claim year	Frequency	Severity	Frequency	Severity
1	0.9216	1.0309	1.1111	1.0309
2	1.0496	1.0347	1.1111	1.0309
3	1.2096	1.0385	1.1111	1.0309
4	1.4096	1.0421	1.1111	1.0309

Concluding Remarks

- Throughout the presentation, we explored ratemaking frameworks that consider longitudinality as well as dependence between frequency and severity components.
- In both cases (where the random effects are either static or dynamic), we could obtain more general formula for credibility premium of the compound sum with dependence between the frequency and severity.
- Use of time-varying random effects processes enables us to account for the seniority of claims in the prediction formulas and lead to many nice features.